

Smooth Hand Preshaping for Visually Realistic Grasping in Virtual Reality

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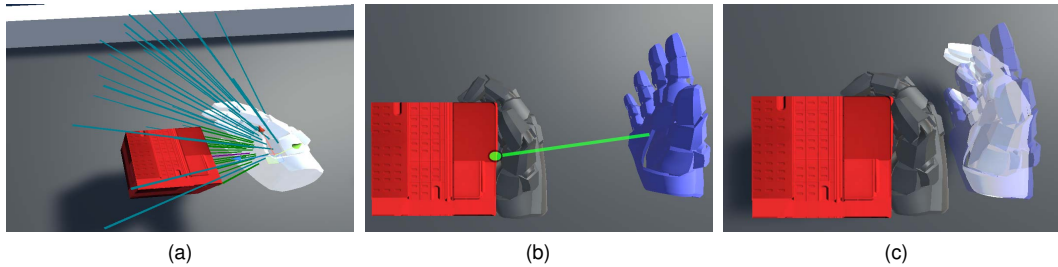


Figure 1: The three steps of our preshaping method (a) Rays are casted from the hand palm to detect a potential object that the user could want to grab (b) We project the hand of the user (in blue) along the selected ray and fit its fingers to the geometry of the object (in black). (c) We only display an interpolated version of the hand according to the distance to the object (in white)

ABSTRACT

In virtual reality, grasping interaction techniques that use controllers often lack a smooth transition between the neutral hand pose and the one that fits the topology of the grabbed object. This transition is referred to as preshaping. We explore the impact on embodiment and task speed of progressively and automatically adapting the fingers' closure of the hand when it gets close to an object by anticipating the user's intention. A comparison with existing methods suggests that this approach negatively impacts the sense of agency but does not impact task speed, body ownership, and interaction and finger movement realism.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual Reality

1 INTRODUCTION

In Virtual Reality (VR), representing controllers with virtual hands is particularly challenging during grasping interactions. Indeed, the positions of the hand and the fingers need to be adapted in order to fit the geometry of the grasped object while avoiding visual interpenetrations. Regarding the three components of a grasping action described by Aydin and Nakajima [1], current methods, in the literature, such as [4] or commercialized, such as Auto-Hand [5], mainly focus on the fitting of the hand to the object. They do not address the problem of preshaping the hand before the grasping action. A hand preshape is the digit posture adopted as the wrist moves towards the object [3]. Current methods often rely on a discrete or continuous button of the controller to grasp an object. Then, the transition between the neutral hand pose and the grasping one is immediate or depends on the speed at which the user presses the continuous input button. It could then be not as smooth as the user would expect.

We then propose an automatic preshaping method that anticipates the grasping intention of the user by progressively and automatically blending the hand pose towards the pose that fits the geometry of the potentially grabbed object. Our research objective is to compare

methods that give full control to the user over the hands and this automatic preshaping approach on task speed, embodiment, and interaction and finger movement realism.

2 AUTOMATIC SMOOTH PRESAPING METHOD

We aim to anticipate grasping actions from the user to produce smoother approaches to objects. The closer to an object the hand gets, the more the fingers positions will shift towards the grasping pose. As shown in Figure 1, our method relies on three steps.

Intention detection As detailed in Figure 1a, we cast rays to detect objects close to the hand palm in a plausible direction for grasping. Rays are sampled in a cone oriented according to the forward vector of the palm and defined by two angles that define its aperture and the maximum distance at which rays are casted. Current used angles values are 120° on the axis that goes from the wrist to the fingers, and 70° on the lateral axis. We use a distance of 15cm and we cast 49 rays per cone. In Figure 1a, the rays that intersect the object are displayed in green. If found any, we weigh the intersection distance of each ray to favor the objects that are closer to the direction of the palm according to the following equation:

$$weight = rawdistance \times (\alpha - \beta \times raydirection \cdot palmdirection)$$

α allows to adjust the impact of the distance to the object on this weight. β allows to adjust the importance of the correspondence between the palm direction and the direction to the object. In our case we use $\alpha = 0.2$ and $\beta = 0.8$. Finally, we select the ray with the lowest weighted distance and the corresponding intersected object. In Figure 1a, the chosen ray is displayed in purple.

Grasping pose estimation We then project the hand on the surface of the identified virtual object to compute a grasping pose that would be relevant if the user would grab it in a very short time. We create a virtual hand placed at the intersection point between the identified ray and the virtual object and with the same orientation as the real hand. We then apply a fitting algorithm in order to bend the fingers of this virtual hand according to the geometry of the virtual object. This process is illustrated in Figure 1b, with the real hand with a neutral pose in blue noted $Hand_{relax}$, the ray in green and the "grasped" version of the hand noted $Hand_{proj}$. The fitting algorithm that we use to adapt the hand to the geometry of a given object is a modified version of the Autohand solution [5], taking inspiration from the work of Oprea et al. [4]. It closes each phalanx progressively with a linear interpolation between a neutral pose and a completely closed one until a collision with the object is detected.

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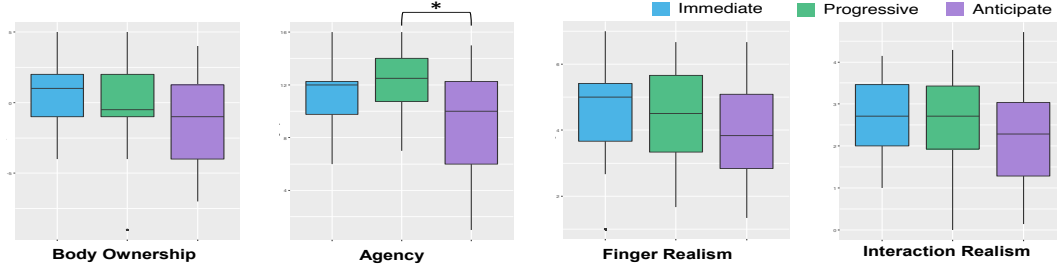


Figure 2: Results for the four subjective criteria we considered: body ownership, agency, interaction and finger realism.

Transition pose computation Finally, we perform a spherical interpolation between the rotation of each joint of each finger in $Hand_{proj}$ and its corresponding one in $Hand_{relax}$. This interpolation is indexed on the distance that separates the hand from the object. It starts when the hand gets within a threshold distance from the object. Here, we used the same value as the raycasting distance: 15cm. In Figure 1c, we can see in black the projected hand, the neutral hand in blue, and the interpolated one displayed to the user in white.

3 EVALUATION

We performed a within-subjects study where we compared this method noted **“Anticipate”** with two existing approaches. In the first one inspired from [5], noted **“Immediate”**, the hand immediately switched from a neutral pose to a pose that fits the object geometry when the users pressed the grip button of the controller. In the second one, inspired from [4], noted **“Progressive”**, the hand pose was linearly blended between the neutral pose and the one that fits the object geometry during a grasping action according to the value of the grip button. For all methods, we used the hand fitting algorithm described in Section 2, and the grabbing action started when pressing the grip button. Releasing the grip button ungrabbed the object. We did not implement two-handed interactions.

Our panel consisted of 24 users, 16 males and 8 females from 20 years old to 57 ($M=39.54$, $SD=13.43$). 21 were right-handed and 3 were left-handed. 14 had prior experience with VR. They were all naive to the experimental hypotheses. Each user was equipped with an Oculus Quest 2 connected to a desktop PC (RTX 2080, Intel Core i5-6600K 3.50GHz, 16Go RAM) with the Air-Link mode.

After a training step, for each preshaping method, each participant performed three different tasks. First, in the *Sorting Task*, on a workbench, the user had to pick and place 15 cubes of different colors (grey, green, blue or red) in a bin with the corresponding color. Second, in the *Conveyor Task*, users were placed in front of a moving conveyor belt. Four types of objects were randomly instantiated (a donut, a tomato, a cube and a triangular prism) on the conveyor moving at a speed of $0.15m/s$. Users had to pick 15 objects and place each of them on its corresponding receptacle on the other side of the conveyor. Last, in the *Accuracy Task*, on a workbench, users had to accurately place 4 objects with complex geometries on their corresponding target displayed with a transparency effect (a “ghost version” of the object). To minimize the ordering effects, the preshaping method was counterbalanced. Questionnaires were fulfilled after performing the three tasks with a given method.

4 RESULTS AND DISCUSSION

Regarding embodiment, we used the questionnaire proposed by Gonzalez-Franco et al. [2] to evaluate the three methods on body ownership and agency. We used the questionnaire proposed by Oprea et al. [4] to evaluate finger movement and interaction realism. Results are detailed in Figure 2. We can observe that our method obtained lower results for the four criteria. We performed a Kruskal-Wallis test for each criterion. For body ownership, and finger and interaction realism we did not observe significant differences

	Immediate	Progressive	Anticipate
Task 1	82.37 ± 37.77	76.25 ± 23.38	81.87 ± 24.48
Task 2	42.23 ± 6.44	43.95 ± 7.49	38.33 ± 5.89
Task 3	61.22 ± 25.92	57.69 ± 21.33	57.39 ± 30.23

Table 1: Time completion in seconds of each task.

($H(2) = 4.49$ $p = 0.11$), ($H(2) = 2.94$ $p = 0.23$), ($H(2) = 2.13$ $p = 0.3452$). Only a significant effect was found for the agency criterion ($H(2) = 8.46$ $p = 0.015$). Dunn’s test with a Bonferroni correction indicated that the sense of agency was significantly lower with our method compared to the Progressive one ($p = 0.011 < 0.05$).

Regarding task speed, the completion times for each task are detailed in Table 1. Only small differences were observed. We performed an ANOVA for the results of each task. For tasks 1 and 3 we did not observe any significant difference ($F(2, 69) = 0.32$ $p = 0.72$), ($F(2, 69) = 0.16$ $p = 0.85$). We did observe a significant difference for the second task ($F(2, 69) = 4.51$ $p = 0.014 < 0.05$). A post-hoc Tukey test highlighted that our method was significantly faster than the progressive method for this task ($p = 0.013 < 0.05$).

The automatic preshaping approach negatively impacted the sense of agency and did not improve the other evaluated criteria. As also reported in open comments, most users disliked the fact that the hands “were moving by themselves”. This study then suggests that when representing controllers with virtual hands, unsolicited obvious movements of these hands should be avoided during grasping tasks. A possible positive impact of our preshaping method on task speed was found when objects were moving and could then be explored in future work. As our method did not rely on real data, we believed that the preshaping behavior was not close enough to reality to improve finger and interaction realism. Using such data as in [6] could be a perspective for future work. Improving intention detection with other data such as gaze tracking could also be considered.

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